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Bihar Engineering University, Patna

B.Tech 2nd Semester Examination, 2026

Time: 03 Hours

Code: 100206 ***Subject: Workshop Practices*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option/answer the following (Any seven question only): **[2×7=14]**

(a) Which of the following tools is primarily used for measuring the thickness of sheet metal accurately?

(i) Scriber (iii) Micrometer
(ii) Divider (iv) Punch

(b) The operation of making holes in a sheet metal using a punch and die is called

(i) Bending (iii) Punching
(ii) Shearing (iv) Folding

(c) Which joining process uses a filler metal having a melting temperature above 450°C?

(i) Soldering (iii) Riveting
(ii) Brazing (iv) Bolting

(d) In arc welding, the heat required for joining is produced by

(i) Chemical reaction (iii) Electric arc
(ii) Friction (iv) Gas flame

(e) Which of the following pattern materials is most commonly used for making patterns in small-scale foundries?

(i) Cast Iron (iii) Wood
(ii) Brass (iv) Wax

(f) The upper part of a moulding box is called

(i) Drag (iii) Cope
(ii) Core (iv) Sprue

(g) Which type of file is commonly used for finishing flat surfaces?

(i) Round File (iii) Flat File
(ii) Half-Round File (iv) Square File

(h) The tool used for making external threads on a rod is called

(i) Tap (iii) Chisel
(ii) Die (iv) Punch

(i) Which woodworking machine is primarily used to reduce the thickness of a wooden board and obtain a smooth surface?

(i) Band Saw (iii) Thickness Planer
(ii) Circular Saw (iv) Wood Working Lathe

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(j) Which forging operation is used to increase the cross-sectional area of a metal workpiece by reducing its length?

(i) Drawing
(ii) Fullering
(iii) Upsetting
(iv) Swaging

Q.2 (a) Explain the various types of sheet metal materials used in workshop practice. Mention their properties and applications. **[7]**

(b) Describe the tools used in sheet metal work with their functions. **[7]**

Q.3 (a) Explain the basic sheet metal operations. **[7]**

(b) Explain the procedure for fabrication of a hexagonal box with a cap using sheet metal. **[7]**

Q.4 (a) Classify the joining processes and explain Brazing and Soldering. **[7]**

(b) Explain Arc Welding and Gas Welding with their advantages and applications. **[7]**

Q.5 (a) Explain the different pattern materials and types of patterns used in foundry practice. **[7]**

(b) Describe the Sand-Casting process with neat sketch. **[7]**

Q.6 (a) Explain the various fitting tools used in a fitting shop and their functions. **[7]**

(b) Describe the classification of files and explain the methods of filing. **[7]**

Q.7 (a) Write the procedures of marking, sawing, and chipping operations in a fitting shop. **[7]**

(b) Explain the types of wood, seasoning of wood and defects in wood. **[7]**

Q.8 (a) Describe various carpentry tools and explain common carpentry joints. **[7]**

(b) Explain the forging tools and equipment and describe various forging operations used in a smithy shop. **[7]**

Q.9 Write short note on any two of the following: **[7×2=14]**

(i) Moulding Sand and its Types
(ii) Safety and Precautions in Fitting Shop
(iii) Safety and Precautions in Carpentry Shop
(iv) Safety and Precautions in Smithy Shop

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